

Qatar-2b

R-0705

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250417 · created 2026-07-27T22:15:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460782.8594 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.226 +/- 0.065
Transit depth (Rp/Rs)^2	5.11 +/- 2.94 [%]
Orbital Inclination (inc)	89.94 +/- 1.74
Airmass coefficient 1 (a1)	0.75 +/- 0.8
Airmass coefficient 2 (a2)	0.32 +/- 1.49
Scatter in the residuals of the lightcurve fit is	138.41 %
Best Comparison Star	#1 - [40, 443]
Optimal Aperture	0.75
Optimal Annulus	3.25
Transit Duration (day)	0.0804 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.226 ± 0.065 vs 0.16327 (deviation 0.96σ)
Duration fit vs published	0.0804 d vs 0.0767 d (deviation 0.79σ)
Mid-time O–C	-4.7 min
Depth SNR	3.4
Residual scatter	138.41 %

Light curve

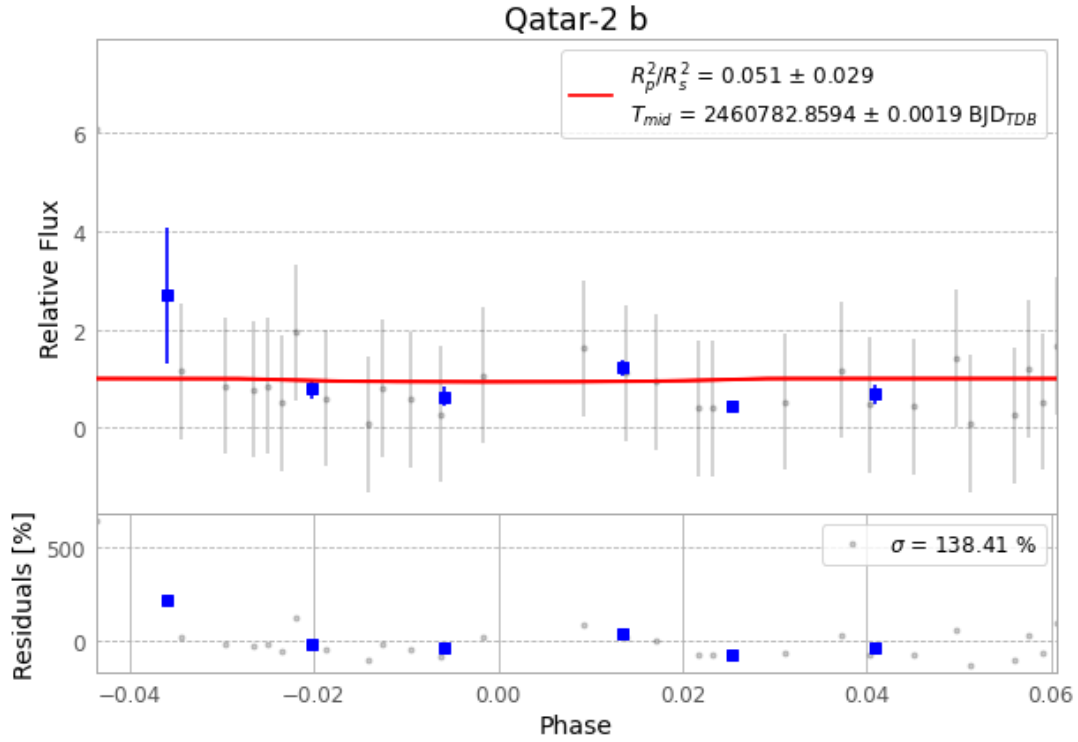


Figure 1 — FinalLightCurve_Qatar-2 b_2025-04-17.png (SHA-256 bd5fb266413770b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 432], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ba511dcbdb3dd095...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

69 FITS frames (45 MB), Qatar-2250417070316.FITS → Qatar-2250417102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250417.log (a167a385fadb...), FinalLightCurve_Qatar-2 b_2025-04-17.png (bd5fb2664137...), FinalLightCurve_Qatar-2 b_2025-04-17.csv (8678751fbed0...)

Compute resources

Wall time 105.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:15Z.